

- 5) Explain the problems of simultaneity as an endogeneity problem.

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13.3 INSTRUMENTAL VARIABLE ESTIMATOR

In this section we discuss about the instrumental variable method of estimation, which is extensively used in empirical research. In instrumental variable method of estimation, the unobservable error (correlated with the explanatory variables) is explicitly allowed. Let us consider the following linear population regression function:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_K x_K + u \quad \dots (13.8)$$

$$E(u) = 0, \quad \text{Cov}(x_j, u) = 0, \quad j = 1, 2, \dots, K-1 \quad \dots (13.9)$$

Let us assume that x_K is correlated with u . In other words, x_K is endogenous and all other explanatory variables $x_1, x_2, \dots, x_{K-1}$ are exogenous in equation (13.8). As mentioned above, endogeneity can come from any of the sources. For the time being let us assume that u contains an omitted variable that is not correlated with any of the explanatory variables $x_1, x_2, \dots, x_{K-1}$, but correlated with x_K . A conditional expectation function can be written as in equation (13.10).

$$E(y | x_1, x_2, \dots, x_K, q) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_K x_K + \gamma q \quad \dots (13.10)$$

In equation (13.10), q is unobservable omitted factor and is correlated with x_K . Equation (13.8) gives us inconsistent estimators of all the β_j under OLS if $\text{Cov}(x_K, u) \neq 0$.

The instrumental variable method provides a way to the endogeneity problem. In IV method we need an instrumental variable z_1 in place of the endogenous variable x_K . In equation (13.8), z_1 should not be present and z_1 should satisfy two conditions:

- (i) The instrumental variable z_1 must not be correlated with u so that z_1 along with $x_1, x_2, \dots, x_{K-1}$ is exogenous in equation (13.8). This can be written as:

$$\text{Cov}(z_1, u) = 0 \quad \dots (13.11)$$

- (ii) The relationship between z_1 and the endogenous variable, x_K is the next requirement. Let the linear projection of x_K onto all the exogenous variables (see Unit 2) be:

$$x_K = \delta_0 + \delta_1 x_1 + \delta_2 x_2 + \dots + \delta_{K-1} x_{K-1} + \theta_1 z_1 + \eta_K \quad \dots (13.12)$$

where, $E(r_K) = 0$ (by definition of a linear projection error)

r_K is uncorrelated with $x_1, x_2, \dots, x_{K-1}$, and z_1 .

This linear projection is based on the assumption that the coefficient of z_1 is non-zero, which can be written as:

$$\theta_1 \neq 0 \quad \dots (13.13)$$

This condition with the statement “both z_1 and x_K are correlated with each other” is not always true. The assumption (13.13) means that z_1 is partially correlated with x_K , once the other exogenous variables $x_1, x_2, \dots, x_{K-1}$ have been netted out. In the case of x_K being the only explanatory variable, the linear projection of x_K can be written as:

$$x_K = \delta_0 + \theta_1 z_1 + r_K \quad \dots (13.14)$$

where,

$$\theta_1 = \text{Cov}(z_1, x_K) / \text{Var}(z_1)$$

$$\theta_1 \neq 0$$

$$\text{Cov}(z_1, x_K) \neq 0$$

Depending upon our requirement, we may put no restrictions on the distribution of x_K or z_1 . Though in many cases x_K and z_1 are continuous, they may be discrete at times. We can also consider one or both of x_K and z_1 as binary variables. These variables may have continuous and discrete characteristics at the same time. In equation (13.12), x_K defines a linear projection, which needs to be defined always in the case of second moments of all variables being finite.

The variable z_1 is considered an instrumental variable for x_K if it satisfies two conditions: (i) variable z_1 is correlated with x_K , and (ii) variable z_1 is not correlated with u . In real life problems, it is difficult to find an instrumental variable.

We can call z_1 as instrument for x_K , because $x_1, x_2, \dots, x_{K-1}$ being uncorrelated with u serve as their own instrumental variables in equation (13.8). For the endogenous explanatory variable x_K the linear projection in equation (13.12) is known as a ‘reduced form equation’. A reduced form always involves describing an endogenous variable as a linear projection onto all exogenous variables in the context of single-equation linear models (we will discuss about reduced form of a structural equation in Unit 16). Using the structural equation (13.8) and the reduced form for x_K , a reduced form for y can be obtained by plugging equation (13.12) into equation (13.8) and re-arranging:

$$y = \alpha_0 + \alpha_1 x_1 + \dots + \alpha_{K-1} x_{K-1} + \lambda_1 z_1 + v \quad \dots (13.15)$$

where, v is the reduced form error

$$v = u + \beta_K r_K$$

$$\alpha_j = \beta_j + \beta_K \delta_j$$

$$\lambda = \beta_K \theta_1$$

Now, OLS can consistently estimate the above reduced form parameters, α_j and λ_1 because by assumption v is uncorrelated with all explanatory variables in equation (13.15). Estimating the structural parameters is more useful although estimates of the reduced form parameters are sometimes of interest.

The assumptions made on the instrumental variable z_1 solve the identification problem for the β_j in equation (13.8). The parameters β_j in terms of population moments in observable variables can be written to indicate identification. Let us write equation (13.8) as:

$$y = x\beta + u \quad \dots (13.16)$$

where $x = (1, x_2, \dots, x_K)$ due to the constant being absorbed into x .

Let us write the $(1 \times K)$ vector of all exogenous variables as

$$z \equiv (1, x_2, \dots, x_{K-1}, z_1)$$

From assumptions (13.8) and (13.9) the K population orthogonality conditions can be written as:

$$E(z'u) = 0 \quad \dots (13.17)$$

Multiplying both sides of equation (13.16) by z' , taking expectations, and using equation (13.18), we get:

$$[E(z'x)]\beta = E(z'y) \quad \dots (13.18)$$

where, $E(z'x)$ is $K \times K$

$E(z'y)$ is $K \times 1$

Here, we observe equation (13.18) as K linear equations with K unknowns $\beta_1, \beta_2, \dots, \beta_K$ which will have unique solution if and only if the $K \times K$ matrix $E(z'x)$ has full rank, which can be written as:

$$\text{rank } E(z'x) = K \quad \dots (13.19)$$

The solution of equation (13.18) can be written as:

$$\beta = [E(z'x)]^{-1}E(z'y) \quad \dots (13.20)$$

Equation (13.20) identifies the vector β where the expectations $E(z'x)$ and $E(z'y)$ can be consistently estimated. We can use random sample (x, y, z_1) to estimate both expectations $E(z'x)$ and $E(z'y)$. The solution (13.20) is valid for condition (13.11), i.e., $\text{Cov}(z_1, u) = 0$. But in order to know whether we used condition (13.13) in deriving solution (13.21), let us assume that exogenous variables are not linearly dependent to each other. Thus, full rank of $E(z'z)$ will be K , which completely rules out perfect collinearity in z in the population. In this case, equation (13.19) will be valid for $\theta_1 \neq 0$. Thus, assumption (13.13) is the major identification condition along with exogeneity condition (13.11) and rank condition (13.19).

Now, the IV estimator of β for a given random sample $\{(x_i, y_i, z_{i1}) : i = 1, 2, \dots, N\}$ from the population can be written as:

$$\hat{\beta} = \left(N^{-1} \sum_{i=1}^N z_1' x_i \right)^{-1} \left(N^{-1} \sum_{i=1}^N z_1' y_i \right) = (Z'X)^{-1} Z'Y \quad \dots (13.21)$$

where, Z : $N \times K$ data matrices

X : $N \times K$ data matrices

Y : $N \times 1$ data vector on the y_i .

The IV estimator is consistent due to the law of large numbers and equation (13.21). In order to identify β , we must give equal importance to conditions (13.11) and (13) though there is a practical difference between them. The difference is condition (13.13) can be tested, whereas condition (13.11) must be maintained because the covariance in condition (11) involves the unobservable u , and therefore we cannot test anything about $Cov(z_1, u) = 0$. Using t-test we can test condition (13.13) in its reduced form (13.12). However, here r_K may not satisfy the homoscedasticity assumption of OLS. Therefore, for $\hat{\theta}_1$, a heteroskedasticity-robust t-statistic is needed. This will be true for x_K being a binary variable or some other variable with discrete characteristics.

Instrumental variable method has a few limitations. It is difficult to find an instrumental variable (z_1) that is perfectly correlated with the endogenous explanatory variable x_k . By using z_1 , we are losing some of the information contained in x_k . This loss of information makes the IV model have larger error terms. Consequently, the standard error of the IV estimator is larger. Therefore, there should be strong correlation between z_1 and x_k . If the correlation is weak, it is better to go look for other instruments, or estimate the model with the endogenous variable and accept the bias.

Check Your Progress 2

- 1) What is meant by instrumental variable in regression analysis?

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- 2) Why is the estimator consistent under instrumental variable method?

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- 3) What is the major difference between the following two conditions while we should give equal importance to both in identifying β ?

- i. $Cov(z_1, u) = 0$
- ii. $\theta_1 \neq 0$

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- 4) Explain the first condition that the instrumental variable must satisfy in order to be considered in the model for estimation.

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13.4 TWO-STAGE LEAST SQUARES ESTIMATOR

There could be several contending instruments for an endogenous variable in a regression model. Which one to choose?

Let us again consider the following two regression models:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_K x_K + u \quad \dots (13.8)$$

$$E(u) = 0, \quad Cov(x_j, u) = 0, \quad j = 1, 2, \dots, K-1 \quad \dots (13.9)$$

where, x_K can be correlated with u . Let us suppose that there is more than one instrumental variable for x_K . Let us also assume that $z_1, z_2, \dots, z_M$ be variables which are not correlated with u so that:

$$Cov(z_h, u) = 0, \quad h = 1, 2, \dots, M \quad \dots (13.22)$$

Equation (13.22) also exhibits that each z_h is exogenous in equation (13.8). There could have M different IV estimators if each z_h has some partial correlation with x_K . Further, we can take a linear combination of these instruments! Thus, apart from the instrumental variables, there could be many more IV because any linear combination of $x_1, x_2, \dots, x_{K-1}, z_1, z_2, \dots, z_M$ is uncorrelated with stochastic error u . In such situations, we follow the method of 'Two-Stage Least Squares (2SLS)'. In 2SLS method we choose that combination of the instruments which is most correlated with the endogenous explanatory variable (x_K).

Under certain assumptions, the two-stage least squares (2SLS) estimator is the most efficient instrumental variable (IV) estimator. To describe the 2SLS method, let us define the vector of exogenous variables given by:

$$z \equiv (1, x_1, x_2, \dots, x_{K-1}, z_1, \dots, z_M) \quad \dots (13.23)$$

where, z is a $1 \times L$ vector and $L = K + M$

In the 2SLS method we choose that linear combination of z (out of many possible linear combinations) which has the maximum correlation with x_K . The linear combination of z having maximum correlation with x_K is given by the linear projection of x_K on z . Now, we can write the reduced form (see Unit 16) of x_K as:

$$x_K = \delta_0 + \delta_1 x_1 + \dots + \delta_{K-1} x_{K-1} + \theta_1 z_1 + \dots + \theta_M z_M + \eta_K \quad \dots (13.24)$$

where, η_K with zero mean is uncorrelated with each of the right-hand-side variables.

As any linear combination of z is uncorrelated with u , we find that x_K^* (being a linear combination of z) is also uncorrelated with u . The linear projection of x_K^* can be written as:

$$x_K^* \equiv \delta_0 + \delta_1 x_1 + \dots + \delta_{K-1} x_{K-1} + \theta_1 z_1 + \dots + \theta_M z_M \quad \dots (13.25)$$

Here, we can interpret x_K^* as a part of x_K , but x_K^* is uncorrelated with u . We can use x_K^* as an instrument for x_K in equation (13.8). However, x_K^* cannot be a usable instrument because θ and δ are population parameters. The parameters in equation (13.24) can be consistently estimated by OLS if there is no multicollinearity problem in the model. The sample analogues of the x_{iK}^* for each observation i , being simply the OLS fitted values, can be written as:

$$\hat{x}_{iK} = \hat{\delta}_0 + \hat{\delta}_1 x_{i1} + \dots + \hat{\delta}_{K-1} x_{i,K-1} + \hat{\theta}_1 z_{i1} + \dots + \hat{\theta}_M z_{iM} \quad \dots (13.26)$$

Defining the vector $\hat{x}_i$ for each observation i , we obtain:

$$\hat{x}_i \equiv (1, x_{i1}, \dots, x_{i,K-1}, \hat{x}_{iK}), \quad i = 1, 2, \dots, N, \quad \dots (13.27)$$

Now, we can use $\hat{x}_i$ as the instruments for x_i . Thus, the IV estimator for the regression model is

$$\hat{\beta} = \left(\sum_{i=1}^N \hat{x}_i' x_i \right)^{-1} \left(\sum_{i=1}^N \hat{x}_i' y_i \right) = (\hat{X}' X)^{-1} \hat{X}' Y \quad \dots (13.28)$$

The IV estimator can be converted into OLS estimator also. We can express $N \times (K+1)$ matrix $\hat{X}$ as:

$$\hat{X} = Z(Z'Z)^{-1} Z'X = P_Z X \quad \dots (13.29)$$

where, P_Z is the projection matrix. Note that $P_Z = Z(Z'Z)^{-1} Z'$ is idempotent and symmetric.

Therefore,

$$\hat{X}' X = X' P_Z X = (P_Z X)' P_Z X = \hat{X}' \hat{X} \quad \dots (13.30)$$

Plugging equation (13.30) into equation (13.28) shows that the IV estimator that uses instruments $\hat{x}_i$ can be written as:

$$\hat{\beta} = (\hat{X}'\hat{X})^{-1}\hat{X}'Y \quad \dots (13.31)$$

The 2SLS method involves application of the least squares method twice. These are:

Step 1: First-Stage Regression

Regress x_K on $1, x_1, \dots, x_{K-1}, z_1, \dots, z_M$ by OLS method Obtain the fitted values of endogenous explanator variable, $\hat{x}_K$.

Step 2: Second-Stage Regression

Regress y on $1, x_1, \dots, x_{K-1}, \hat{x}_K$ by OLS method. Obtain 2SLS estimators for all the parameters.

Note that both 2SLS and IV estimators will be the same if we have only one instrument for x_K .

Check Your Progress 3

- 1) Write down the reduced form for x_K .

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- 2) Write down the linear projection of x_K^* .

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- 3) Write down the IV estimator using $\hat{x}_i$ as the instruments for x_i .

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- 4) Explain the steps followed in two-stage least squares method to obtain $\hat{\beta}$.

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- 5) Describe the importance of the 2SLS method.

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13.5 LET US SUM UP

An assumption of the classical regression model is that the error term is independent of the explanatory variables in the model. In other words, there is no correlation between the error term and the explanatory variables. If this assumption is violated, we cannot get best linear unbiased estimators of the parameters. If any of the explanatory variables is correlated with the error term we face the endogeneity problem.

There are three possible sources of endogeneity problem in applied econometrics, viz., omitted variables, measurement error, and simultaneity. The problem of omitted variables arises when we are unable to include some variables in a regression model due to data unavailability. Errors-in-variables can occur due to various reasons such as nonresponse error, reporting error and computing error. Measurement error in explanatory variables also creates an endogeneity bias. If any of the explanatory variables is determined simultaneously along with the dependent variable, then the problem of simultaneity arises. Sharp distinction among the three possible forms of endogeneity is not possible always. We in fact observe more than one source of endogeneity in any given equation.

For solving the problem of endogeneity, we have two methods: (i) instrumental variable method, and (ii) two-stage least squares method (2SLS). In IV estimation we replace the endogenous explanatory variable with an instrumental variable. An instrumental variable should have two properties: (i) there is a strong correlation between the endogenous explanatory variable (x_k) and instrument under consideration, and (ii) the instrumental variable is not correlated with the stochastic error term (u).

There could be more than one instrumental variable for an endogenous explanatory variable. In such situations, it is difficult to make a choice among the instrumental variables. We apply the method of two-stage least squares (2SLS) in such situations. We take all possible instruments of the endogenous variable x_k , while avoiding the problem of multicollinearity among the instruments. In the first stage, we run OLS of x_k on all instrumental variables and obtain $\hat{x}_k$. In the second stage, we run OLS of y on all the explanatory variables; while replacing x_k by $\hat{x}_k$. Both 2SLS and IV estimators will be same if we have only one instrument for x_k .

13.6 KEY WORDS

Endogeneity Problem : In an econometric model if any of the explanatory variables is correlated with the stochastic error term, the concerned explanatory variable is said to be endogenous in nature.

Measurement Error : The observed variables on which we can collect the data is modelled as $\tilde{x} = x + u$ and $\tilde{y} = y + v$, which show that our observed variables $\tilde{x}$ and $\tilde{y}$ are measured with additive errors u and v respectively. Thus u and v are measurement errors.

Instrumental Variables : A set of variables known as instrumental variables are identified with the condition that they are correlated with the independent variables but uncorrelated at least asymptotically with composite disturbances so that we can get consistent estimator.

Simultaneity : If at least one of the explanatory variables is determined simultaneously along with y then the problem of simultaneity arises. Say for example, x_K and u are correlated if x_K is determined partly as a function of y . There are always no sharp distinctions among the three possible forms of endogeneity. We in fact observe more than one source of endogeneity in any given equation.

13.7 SOME USEFUL BOOKS AND REFERENCES

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